



Technical Service Bulletin

Technical Service Bulletin: TSB101103

Released Date: 07-Jan-2011

ISX Cylinder Block Compatibility

ISX Cylinder Block Compatibility

Warranty Statement

The information in this document has no effect on present warranty coverage or repair practices, nor does it authorize TRP or Campaign actions.

Contents

The purpose of this document is to notify the field of a new ISX 15 Liter, backward compatible, 2010 engine block. The ISX engine block has been modified for use on ISX15 CM2250 engines to accommodate changes needed to meet EPA 2010 emission level standards.

Similar to the ISX 2007 engine block, the new ISX15 2010 engine block, Part Number 3686832, is backward compatible with **all** ISX products.

- The Cylinder Block Kit, Part Number 2881765, contains the parts that are required for any ISX model year engine block replacement. This kit contains a dressed cylinder block, Part Number 3686832, that includes cylinder liners and main caps, as well as a cylinder block hardware kit, Part Number 4955501.

Cylinder Block Kit; Part Number 2881765

New Part Number	Description
3686832	Cylinder Block
4955501	Cylinder Block Hardware Kit

Cylinder Block Hardware Kit; Part Number 4955501

New Part Number	Quantity	Description
3412860	6	Piston Cooling Nozzles
3679037	6	Piston Cooling Nozzle Capscrews
3678610	2	Threaded Plugs
3678921	5	Threaded Plugs
3679334	3	Expansion Plugs
3007636	1	Expansion Plugs

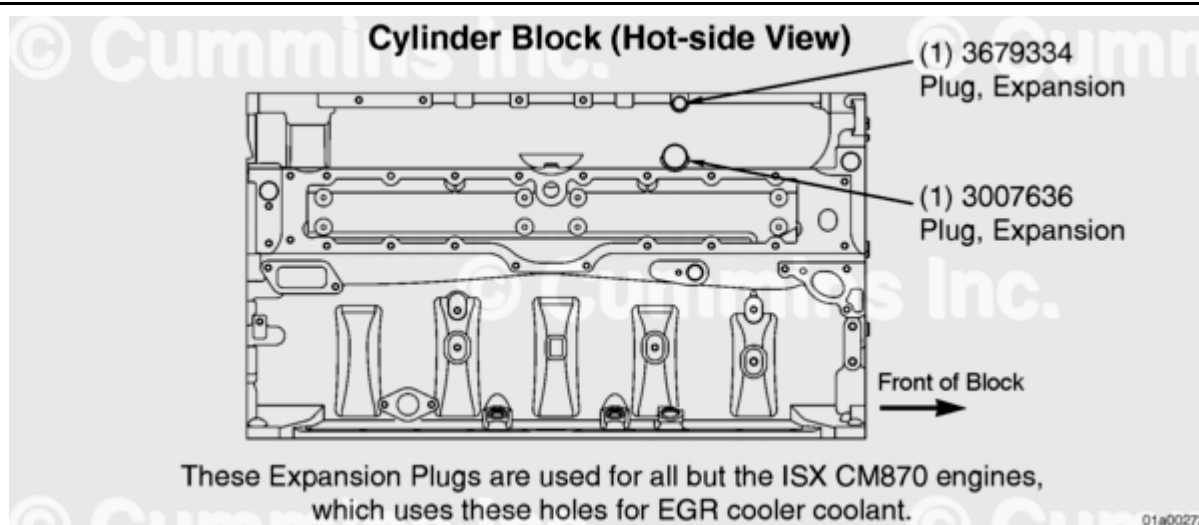
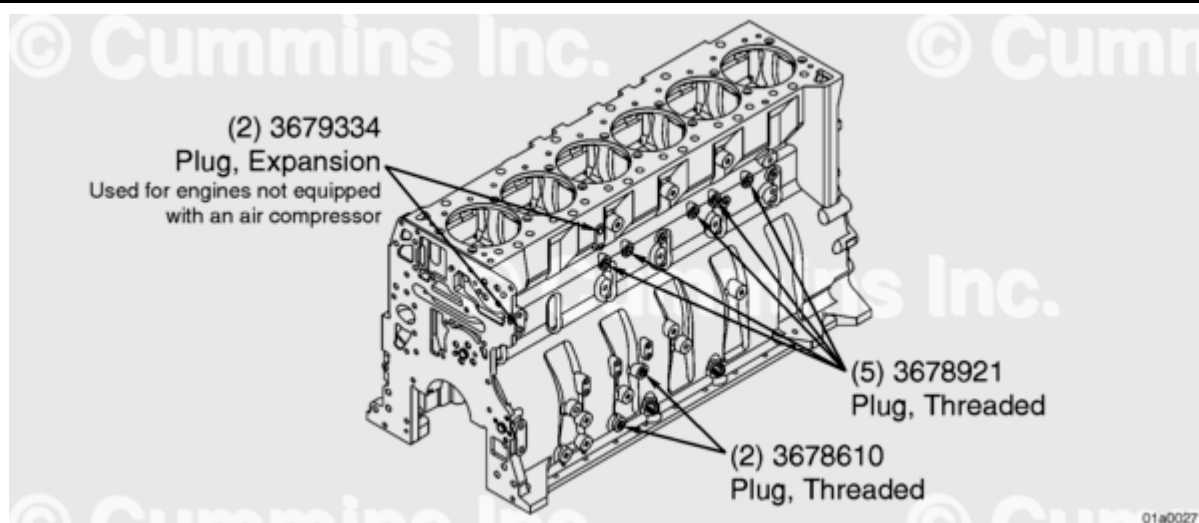
Cylinder Block Hardware Kit; Part Number 4955501

New Part Number	Quantity	Description
3678871	4	Ring Dowels
3900257	4	Dowel Pins

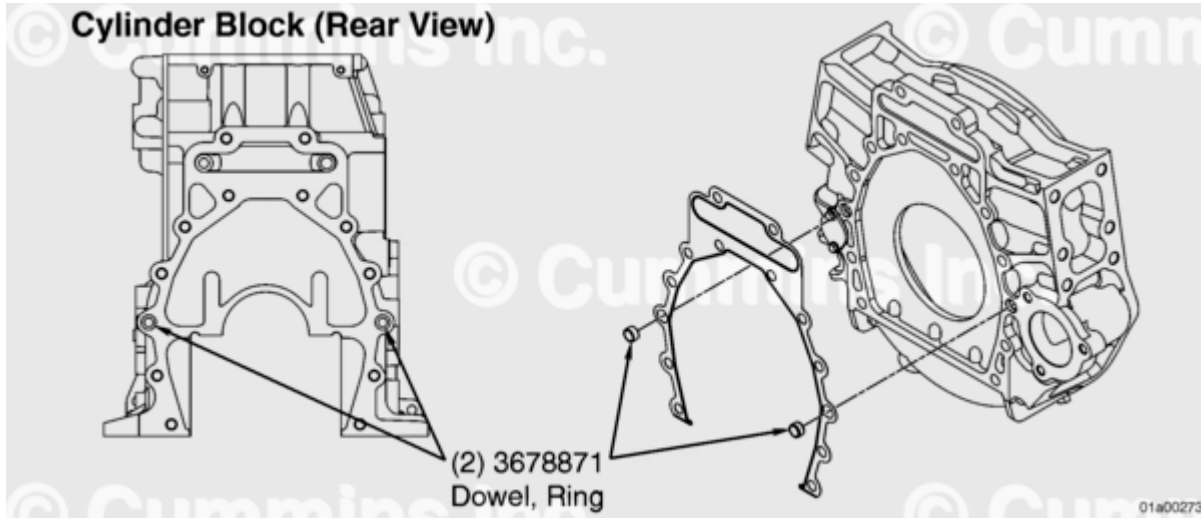
Not all of the components included in the Cylinder Block Hardware Kit, Part Number 4955501 will be needed for every ISX engine model.

Additional engine model specific parts will be needed for engine block replacement.

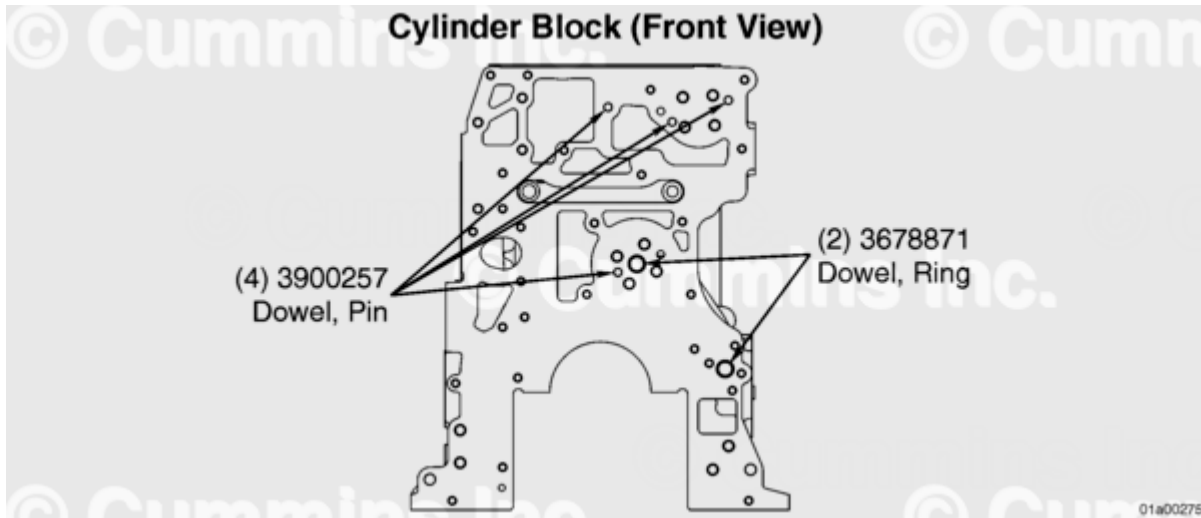
Cylinder Block Hardware Kit; Part Number 4955501, Location Breakdown



Cylinder Block (Rear View)



Cylinder Block (Front View)



Additional Parts Not Included in Part Number 2881765 Kit:

In addition to installing the components included in Cylinder Block Kit, Part Number 2881765, engine model specific parts are needed during a block replacement as listed below.

The ISX15 CM2250 cylinder blocks use a unique gear housing gasket and cylinder head gasket, and are compatible to this service block by virtue of installing:

New Part Number	Description
3685510	ISX15 CM2250 Gear Housing Gasket
3685639	ISX15 CM2250 Head Gasket

The parts listed above **must** be used for **any** service event, when these components need to be replaced. It is critical that these parts are used. The ISX CM871 and older gaskets appear similar and fit; however, use of the CM871 and older parts are likely to cause damage to the engine.

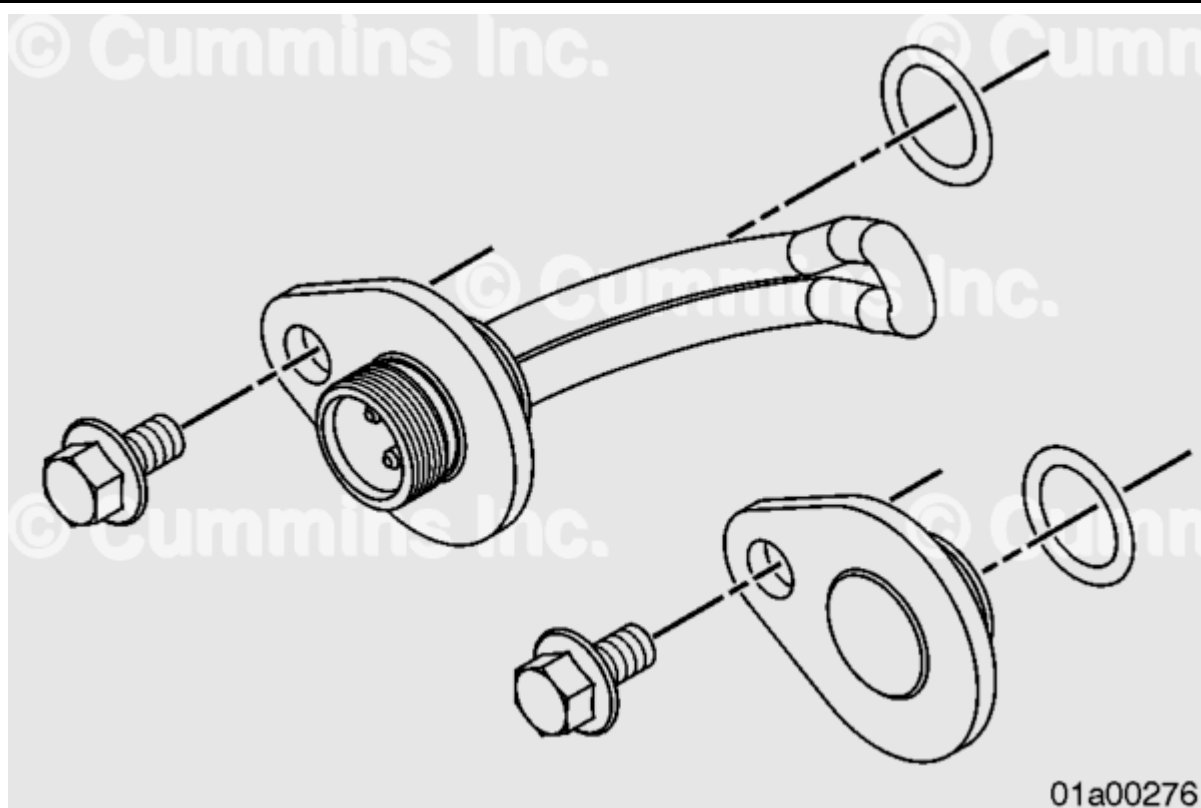
The ISX CM871, CM870, CM570, and Signature™ cylinder blocks are compatible with the new blocks by virtue of installing:

New Part Number	Description
4985562	CM871, CM870, CM570, Signature™ Gear Housing Gasket
3685834	CM871, CM870, CM570, Signature™ Head Gasket

ESN prior to 79259454 (Built July 24, 2007):

The new cylinder block features the ISX CM871 designed block heater port and requires the use of a CM871 designed block heater or a CM871 designed block heater port plug. The new cylinder block is backward compatible to cylinder blocks with an ESN prior to 79259454, by virtue of installing a new block heater or block heater port plug.

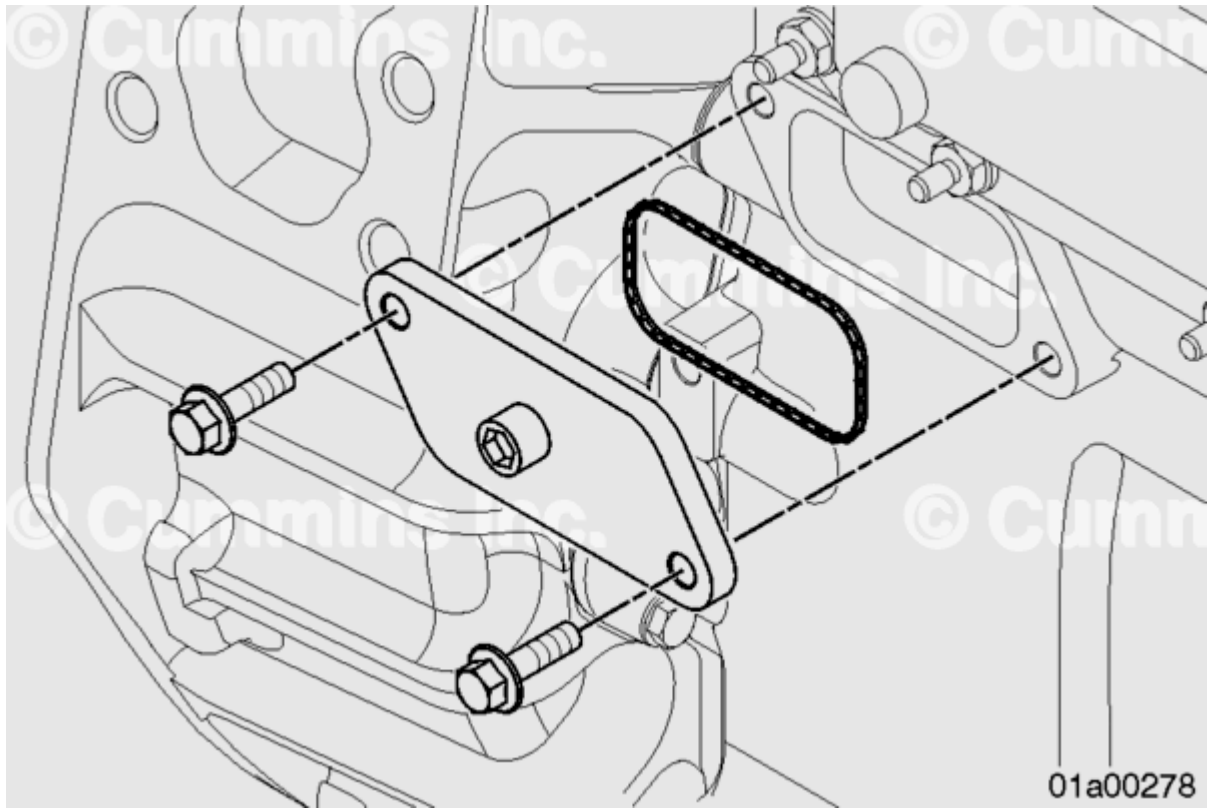
Reference TSB100758 for additional information and part numbers for the block heater and block heater port plug.



ISX CM870, CM570, Signature™ (ESN prior to 79204200, built August 31, 2006):

The new cylinder block features a coolant return port that was specifically designed to provide a coolant return for the ISX CM2250 and CM871 EGR coolers. The coolant port is located on the exhaust side, at the rear of the engine, just below the lubricating oil cooler. The new cylinder block is backward compatible to ISX CM870 and older cylinder blocks, by virtue of installing a coolant cover plate.

Reference TSB100526 for additional information and part numbers for the coolant cover plate.



Early Signature™ Engines

Approximately the first 950 ISX engines built, ESN 14000000 through 14000950, used a different tone wheel on the crankshaft than the crankshaft now in production, and can **not** be replaced with this service block kit.

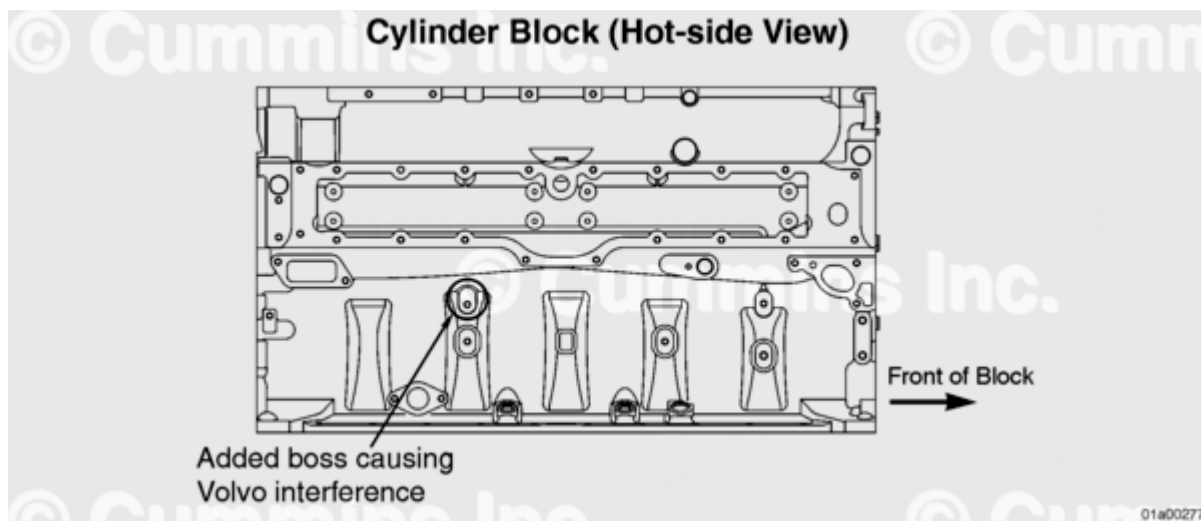
Reference TSB100331 for additional information on replacing the blocks for these engines.

Chassis Specific Considerations (Volvo)

For Volvo VNL Chassis using engines prior to the ISX15 CM2250, the power steering coolant supply line may interfere with a new boss, added to the block in this kit, Part Number 2881765. In order to use the original coolant lines with this service block, the interfering boss **must** be ground down to provide clearance.

Use protective plug kit, Part Number 4918319, to plug open ports and tape off any additional open ports on the cylinder block before grinding to avoid debris entering the engine during the grinding process. Thoroughly clean and inspect the block before assembling the engine.

Note : Use caution not to damage the block. Do not grind the boss deeper than the threaded capscrew hole inside of the boss.



Warranty Statement

The information in this document has no effect on current warranty coverage or repair practices, nor does it authorize TRP or Campaign actions.

Document History

Date	Details
2010-11-24	Module Created

Last Modified: 07-Jan-2011
